

OEAS 705/805 Advanced Environmental Data Science

Fall 2026

Week 5

Dr. Jessie Turner, jturners@odu.edu

Class times: OCNPS 320, T/Th, 11:00 - 12:15

Office hours: After class T/Th,

And: OCNPS 411, M, 1:00 - 3:00



Schedule

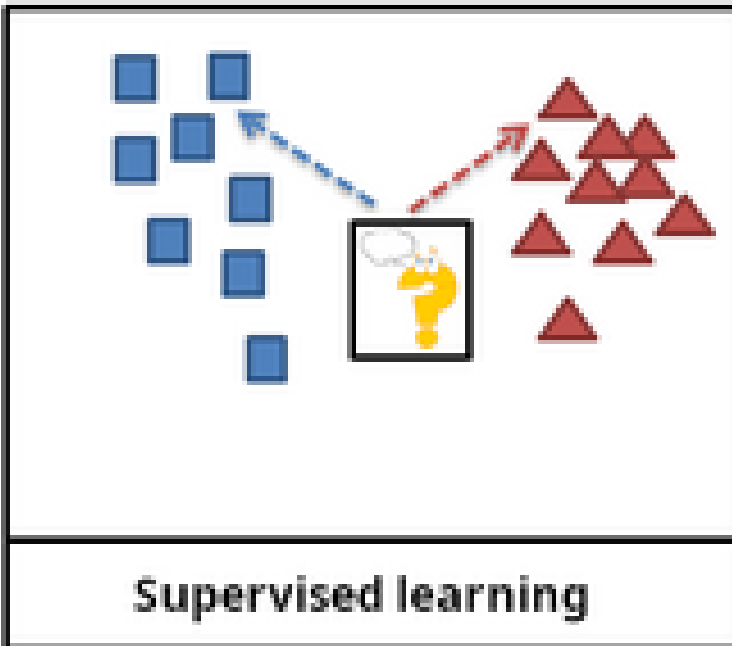
No Class Tuesday October 13th – Fall Break
Resume Thursday October 15th

Homework 3 Due Monday Oct 19th
→ Post link to your OEAS805/HW3 GitHub to Canvas

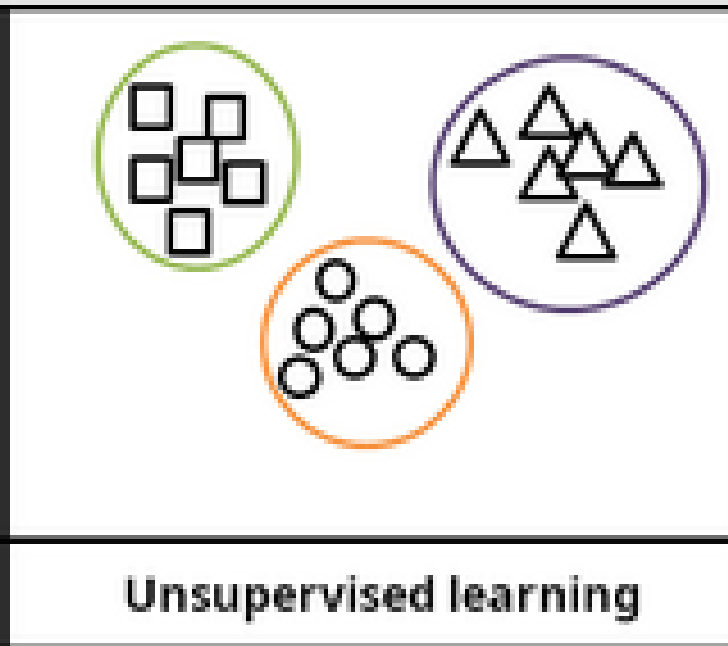
Projects complete November 23rd for Peer Review
Peer Reviews Due by November 30th (graded for completion only)

Machine Learning

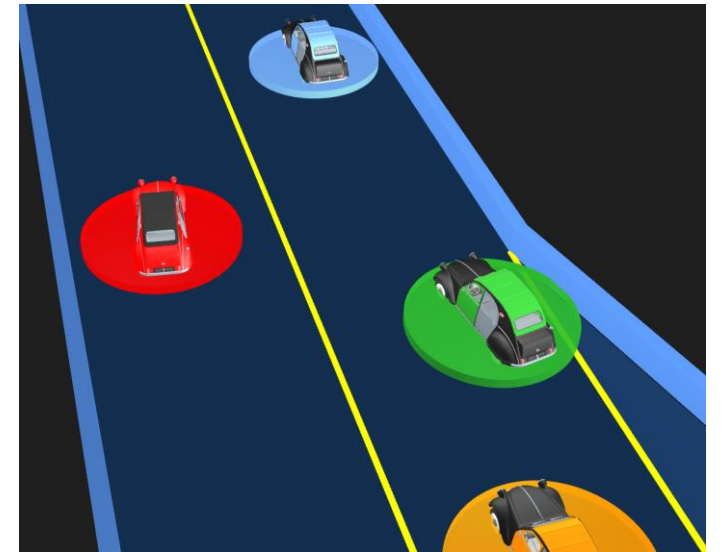
1. Supervised Learning



2. Unsupervised Learning



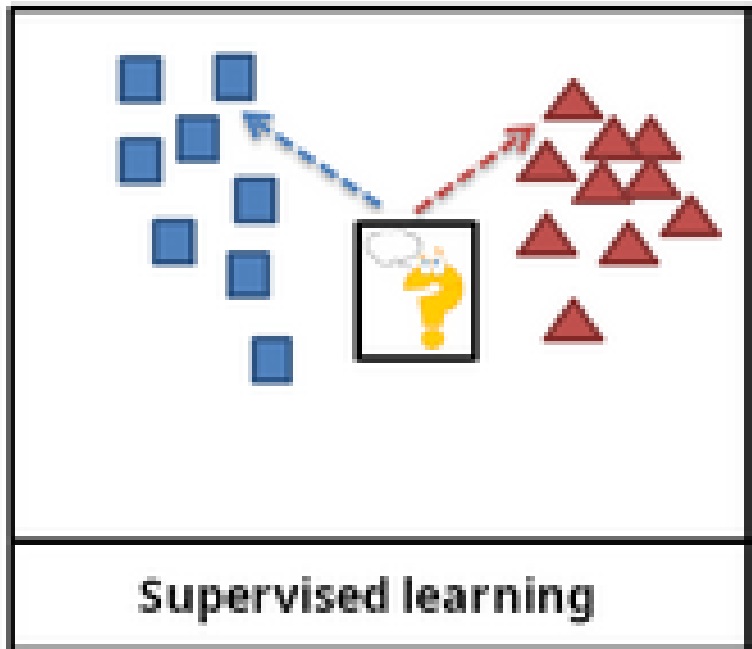
3. Reinforcement Learning



Game Theory applications,
Not as useful in earth sciences

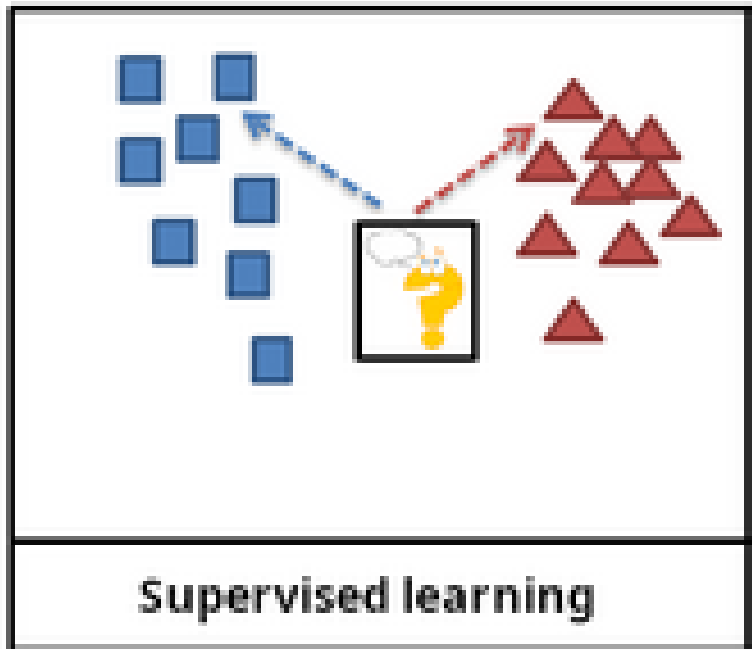
Machine Learning

1. Supervised Learning



Machine Learning

1. Supervised Learning

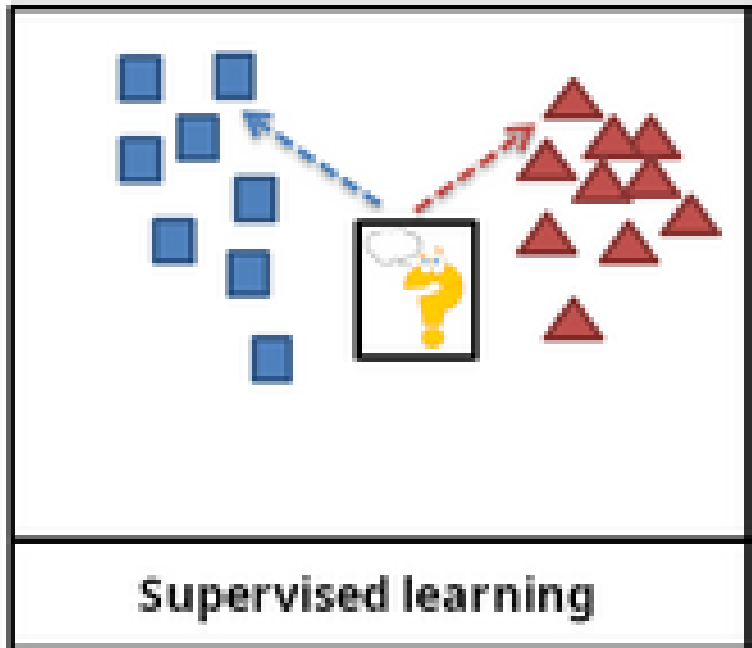


Labeled samples are used to train the model

- Regression
- Classification

Machine Learning

1. Supervised Learning

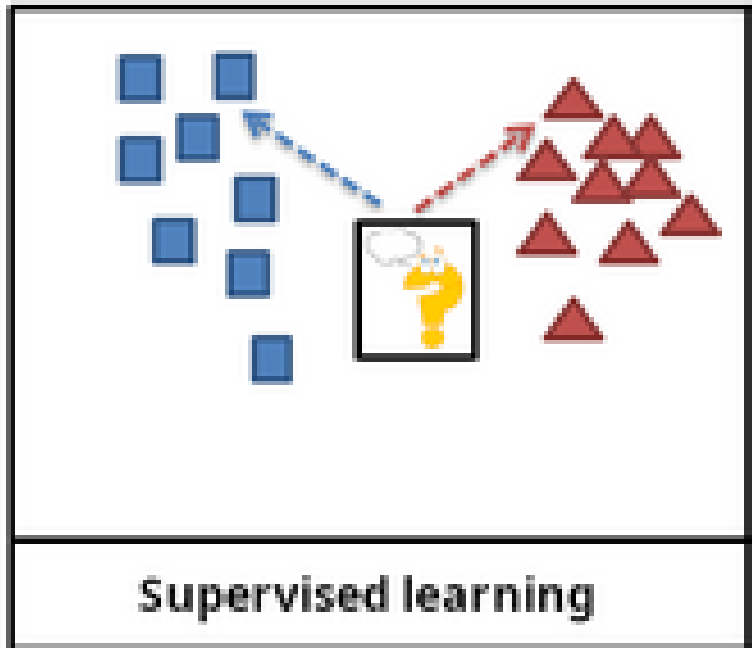


Labeled samples are used to train the model

- Regression – predict values
- Classification – predict category

Machine Learning

1. Supervised Learning

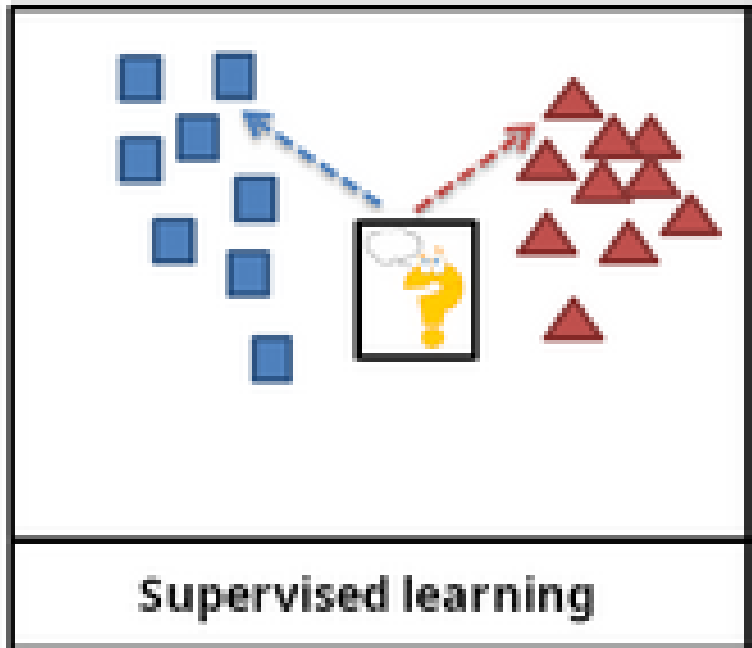


Labeled samples are used to train the model

- **Regression – predict values**
- Classification – predict category

Machine Learning

1. Supervised Learning

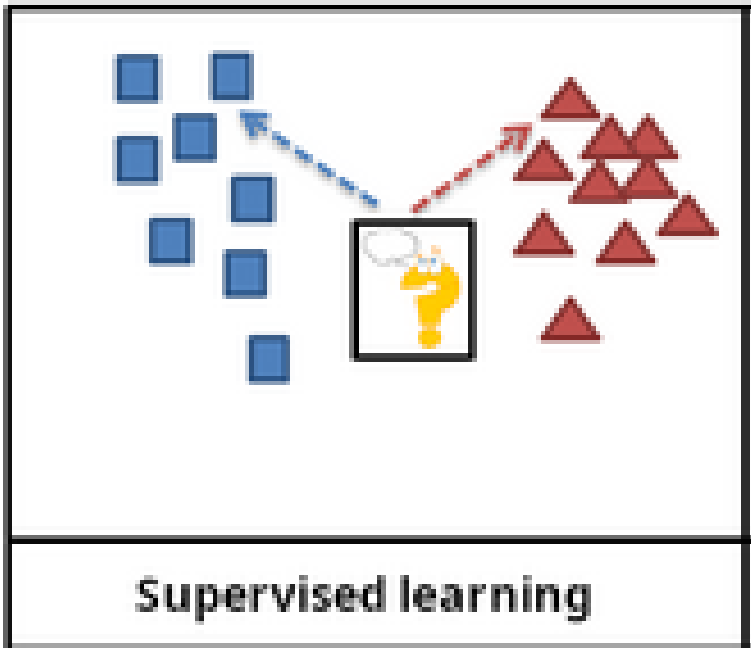


Examples:

- Random forests
- Support vector machines
- Gaussian processes
- Neural networks

Machine Learning

1. Supervised Learning



Examples:

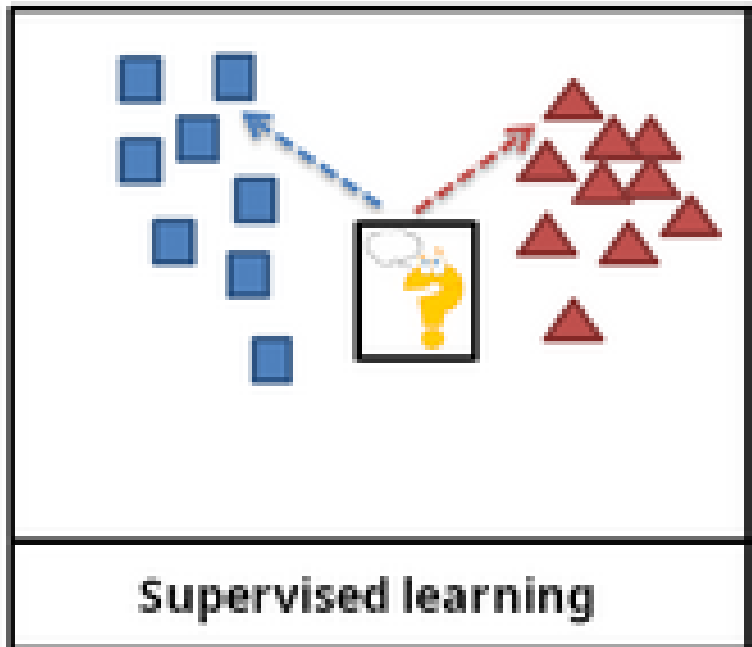
- **Random forests**
- Support vector machines
- **Gaussian processes**
- Neural networks

We will try these today

Developing Supervised ML Models

1. Supervised Learning

1. Training
2. Validation
3. Testing



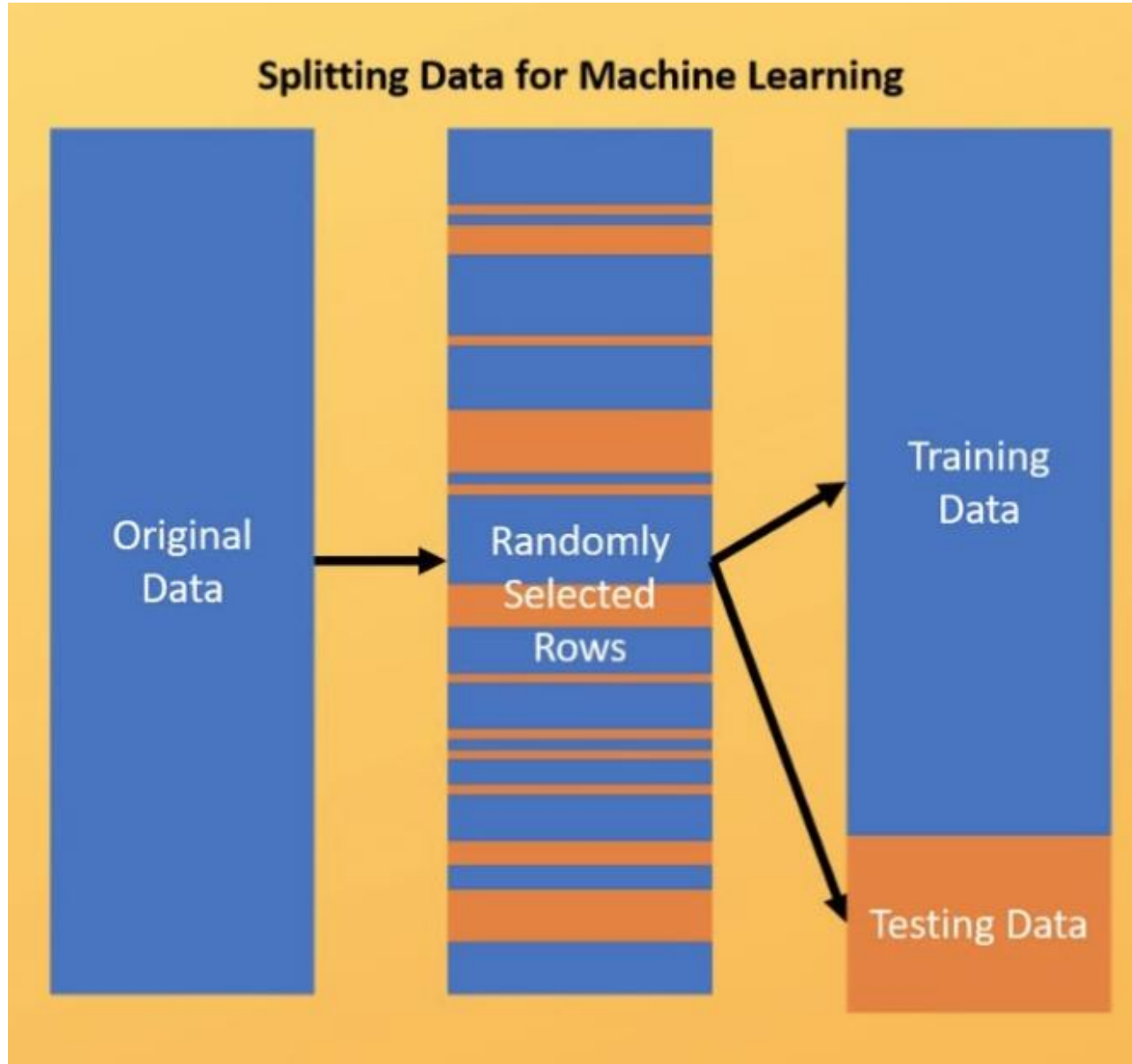
Developing Supervised ML Models

Sometimes Training and Validation are grouped together:

- | | | |
|-------------------|--|-------------------|
| 1. Training |  | 1. Training |
| 2. Validation | | 2. Testing |
| 3. Testing | | |

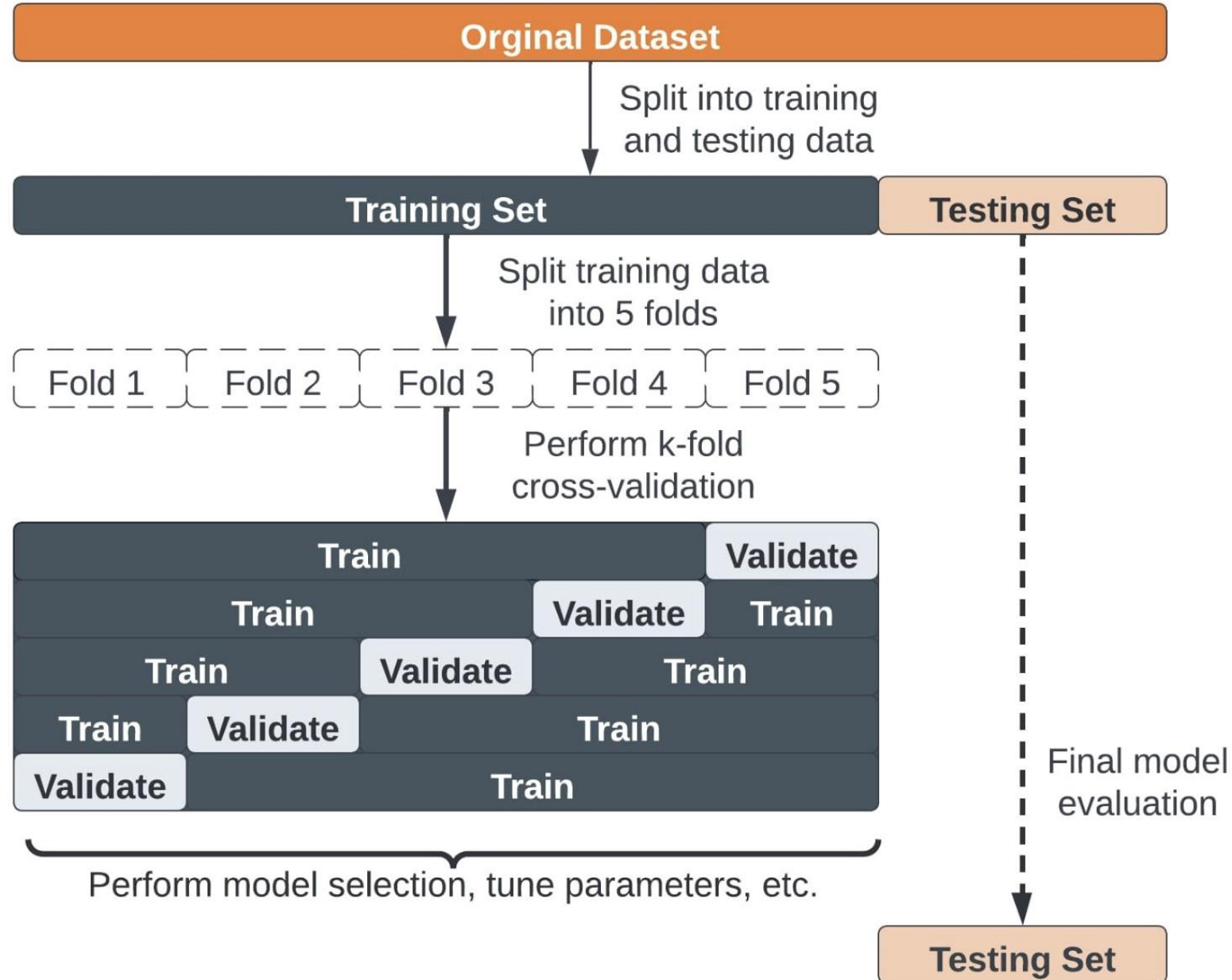
**model's performance is tested on
previously unseen data**

Developing Supervised ML Models



*Set it up so that randomly selected rows are selected the same way every time!

Developing Supervised ML Models



Supervised ML Models for *Regression* that we will try out:

regression_examples.ipynb

- Linear Regressions (ordinary least squares)
- K-Nearest-Neighbors (KNN) Regressions
 - Linear
 - Non-linear
- Multi-Layer Perceptron Regressions
 - Linear
 - Non-linear
- Radius Neighbors Regressions
 - Linear
 - Non-linear

Supervised ML Models for *Regression* that we will try out:

Create a fresh jupyter notebook

One half of the room:

- Random Forest

Other half of the room:

- Gaussian Process

Consider:

- *What can be tuned?*
- *Advantage over other regression methods?*

Supervised ML Models for *Regression* that we will try out:

Pairs, each pair:

One person who just did random forest

One person who just did gaussian process

*Teach the other person
what you just did.*